General Bus Motor Protocol Specification

Communication method: This protocol uses a single-wire, half-duplex asynchronous serial bus or an RS485 bus. Each frame consists of 1 start bit, 8 data bits, and 1 stop bit, without a parity bit, for a total of 10 bits.
The controller and motor communicate using a request–response mechanism: the controller sends an instruction packet and the motor returns a response packet.
ID: Each motor on the same bus has its own ID. IDs range from 0 to 253, corresponding to hexadecimal 0x00–0xFD.
The instruction packet sent by the controller contains an ID. Only a motor with a matching ID can accept the instruction and send the corresponding response.

Instruction Packet

Packet structure:	0xFF	0xFF	ID	Length	Instruction	Parameter 1	…	Parameter N	CheckSum
0xFF 0xFF	Two 0xFF bytes indicate the instruction-packet header.								
ID	Unique ID for each motor. Up to 254 motors: 0x00–0xFD.								
	0xFE is the broadcast ID; packets addressed to it control all motors on the connected network.								
	Broadcast packets do not produce returned status packets.								
Length	Packet length = number of transmitted parameter bytes + 2.								
Instruction	Instruction executed by the motor.								
Parameter 1 ~ N	Transmitted data bytes.								
CheckSum	Packet checksum, calculated as: CheckSum = ~(ID + Length + Instruction + Parameter 1 + … + Parameter N)								

Response Packet

Packet structure:	0xFF	0xFF	ID	Length	Status	Pre Len	Instruction	Pre Para 1	Para 1	…	Para N	CheckSum
0xFF 0xFF	Two 0xFF bytes indicate the response-packet header.											
ID	ID of the responding motor.											
Length	Response packet length = number of received parameter bytes + 2.											
Status	Current motor status (see Motor Status below).											
Pre Len	Length of the instruction packet.											
Instruction	Instruction executed by the motor.											
Pre Para 1	For Read, Write, or Reg Write, this is Parameter 1 of the instruction packet; for other instructions, 0.											
Para 1 ~ N	Response data bytes.											
CheckSum	Packet checksum, calculated as: CheckSum = ~（ID + Length + Status + Length2 + Instruction + Pre Para 1 + Parameter 1 + … + Parameter N）											

Motor Status (Status)

Bit	Name	Description
Bit 7	——	
Bit 6	Parsing error	Set to 1 if the packet format violates the protocol and cannot be parsed.
Bit 5	Stall error	Set to 1 if the motor stalls.
Bit 4	Checksum error	Set to 1 if the instruction packet checksum is incorrect.
Bit 3	Length/ range error	Set to 1 if the instruction packet length exceeds the permitted length.
Bit 2	Temperature error	Set to 1 if internal motor temperature exceeds the configured upper limit.
Bit 1	Angle error	Set to 1 if target position exceeds configured minimum/maximum limits.
Bit 0	Voltage error	Set to 1 if input voltage falls outside the configured operating voltage range.

Example: The motor returns the packet: 0xFF 0xFF 0x01 0x02 0x24 0xD8
This means motor ID 1 has status 0x24. In binary, 0x24 is 00100100; bits 5 and 2 are set, indicating a motor stall error and a temperature error.

Instructions

Instruction	Function	Value	Parameter length
Ping	Retrieve a status packet.	0x01	0x00
Read Data	Read data from the control table.	0x02	0x02
Write Data	Write data to the control table.	0x03	N + 2 (N: number of parameters)
Reg Write	Queue a write; do not execute until an Action instruction is received.	0x04	N + 2 (N: number of parameters)
Action	Execute commands queued by Reg Write.	0x05	0
Reset	Restore motor control-table values to factory defaults.	0x06	0
Sync Write	Control multiple motors simultaneously.	0x83	(L + 1) × N + 4 (L: data bytes per motor; N: number of motors)

Ping Instruction
Function: Query the current motor status.
Length: 0x02
Instruction: 0x01
Parameters: None
Example:
Send a Ping instruction to motor ID 1.
Instruction packet: 0xFF 0xFF 0x01 0x02 0x01 0xFB

Header	ID	Data length	Instruction	Parameters	Checksum
0xFF 0xFF	0x01	0x02	0x01	None	0xFB

Response packet: 0xFF 0xFF 0x01 0x05 0x00 0x02 0x01 0x00 0xF6

Header	ID	Data length	Status	Previous length	Instruction	Previous parameter 1	Parameters	Checksum
0xFF 0xFF	0x01	0x05	0x00	0x02	0x01	0x00	None	0xF6

Read Data Instruction
Function: Read data from the motor control table.
Length: 0x04
Instruction: 0x02
Parameter 1: Starting address of the data to read.
Parameter 2: Number of bytes to read.
Example:
Read the current temperature of motor ID 1.
Parameters: 0x2B 0x01, meaning read one byte from address 0x2B.
Instruction packet: 0xFF 0xFF 0x01 0x04 0x02 0x2B 0x01 0xCC

Header	ID	Data length	Instruction	Parameters	Checksum
0xFF 0xFF	0x01	0x04	0x02	0x2B 0x01	0xCC

Response packet: 0xFF 0xFF 0x01 0x06 0x00 0x04 0x02 0x2B 0x1D 0xAA

Header	ID	Data length	Status	Previous length	Instruction	Previous parameter 1	Parameters	Checksum
0xFF 0xFF	0x01	0x06	0x00	0x04	0x02	0x2B	0x1D	0xAA

The returned parameter is 0x1D, indicating a current temperature of 29°C.

Write Data Instruction
Function: Write data to the motor control table.
Length: N + 2 (N: number of parameters).
Instruction: 0x03
Parameter 1: Starting address for writing data.
Parameter 2: First data byte to write.
Parameter 3: Second data byte to write.
Parameter N+1: Nth data byte to write.
Example:
Use a broadcast instruction to change a motor with any ID to ID 1.

Parameters: 0x03 0x01, meaning write 0x01 (ID 1) to address 0x03.
Instruction packet: 0xFF 0xFF 0xFE 0x04 0x03 0x03 0x01 0xF6

Header	ID	Data length	Instruction	Parameters	Checksum
0xFF 0xFF	0xFE	0x04	0x03	0x03 0x01	0xF6

Because the instruction is broadcast, no response packet is returned.

Reg Write (Deferred Write) Instruction
Function: Reg Write is similar to Write Data, but differs in when it executes. On receiving a Reg Write packet, the motor stores the received data in a buffer and sets the registered-instruction flag (address 0x2C) to 1. Only after receiving an Action instruction does the motor write the buffered data to its control table.
Length: N + 2 (N: number of parameters).
Instruction: 0x04
Parameter 1: Starting address for writing data.
Parameter 2: First data byte to write.
Parameter 3: Second data byte to write.
Parameter N+1: Nth data byte to write.
Example:
Move motor ID 1 to position 500.
Parameters: 0x1E 0xF4 0x01; write to address 0x1E with position 500 (0x1F4).
Instruction packet: 0xFF 0xFF 0x01 0x05 0x04 0x1E 0xF4 0x01 0xE2

Header	ID	Data length	Instruction	Parameters	Checksum
0xFF 0xFF	0x01	0x05	0x04	0x1E 0xF4 0x01	0xE2

Response packet: 0xFF 0xFF 0x01 0x05 0x00 0x05 0x04 0x1e 0xD2

Header	ID	Data length	Status	Previous length	Instruction	Previous parameter 1	Parameters	Checksum
0xFF 0xFF	0x01	0x05	0x00	0x05	0x04	0x1E	None	0xD2

Action (Execute Deferred Write) Instruction
Function: Write data buffered by a Reg Write instruction into the control table and set the registered-instruction flag (address 0x2C) to 0.
Length: 0x02
Instruction: 0x05
Parameters: None
The Action instruction is generally used when multiple motors must be controlled synchronously.
When controlling multiple motors using Write Data, there is a time difference between when the first and last motor receive their instructions. Instead, send data to each motor using Reg Write, then broadcast an Action instruction using ID 0xFE. All motors will receive and execute the Action instruction simultaneously.
Example:
Broadcast an instruction to execute on all motors.
Instruction packet: 0xFF 0xFF 0xFE 0x02 0x05 0xFA

Header	ID	Data length	Instruction	Parameters	Checksum
0xFF 0xFF	0xFE	0x02	0x05	None	0xFA

Because the instruction is broadcast, no response packet is returned.

Sync Write (Synchronous Write) Instruction
Function: Write data to the control tables of multiple motors simultaneously. The broadcast ID must be used.
ID: 0xFE
Length: (L + 1) × N + 4 (L: length of data for each motor; N: number of motors).
Instruction: 0x83
Parameter 1: Starting address for writing data.
Parameter 2: Length of data to write (L).
Parameter 3: ID of the first motor.
Parameter 4: First data byte for the first motor.
Parameter 5: Second data byte for the first motor.
…
Parameter L+3: Lth data byte for the first motor.
Parameter L+4: First data byte for the second motor.
Parameter L+5: Second data byte for the second motor.
…
Parameter 2L+4: Lth data byte for the second motor.

Sync Write is also used to control multiple motors simultaneously.
A single Sync Write instruction writes to the control tables of multiple motors in one operation, whereas Reg Write + Action accomplishes this in stages. Sync Write therefore offers better real-time performance and shorter communication time. However, the data length and starting address must be the same for all motors.

Example:
Write the following positions and speeds to four motors simultaneously:
ID 0: Position: 0X010; Speed: 0X150
ID 1: Position: 0X220; Speed: 0X360
ID 2: Position: 0X030; Speed: 0X170
ID 3: Position: 0X220; Speed: 0X380

Instruction packet: 0XFF 0XFF 0XFE 0X18 0X83 0X1E 0X04 0X00 0X10 0X00 0X50 0X01 0X01 0X20 0X02 0X60 0X03 0X02 0X30 0X00 0X70 0X01 0X03 0X20 0X02 0X80 0X03 0X12

Header	ID	Data length	Instruction	Parameters
0xFF 0xFF	0xFE	0x18	0x83	0x1E 0x04 0X00 0X10 0X00 0X50 0X01 0X01 0X20 0X02 0X60 0X03 0X02 0X30 0X00 0X70 0X01 0X03 0X20 0X02 0X80 0X03

Because the instruction is broadcast, no response packet is returned.

Control Table

Memory Regions (Flash, RAM)
The control table is divided into two parts. Values in the RAM region are initialized whenever the motor powers on. The Flash region retains user-modified values after power-off and restores them at the next power-on.

Read/Write
The control table has two access types: RW means readable and writable; R means read-only, so write instructions have no effect. Read-only (R) values are generally used to measure and monitor motor status, while RW values are generally used for motor control.

Default Values
In the Flash region, the default value is the factory default. In the RAM region, it is the value assigned by default each time the motor powers on.

High and Low Bytes
Some control-table addresses share a name with H and L suffixes. These are 16-bit values split between a high-byte address and a low-byte address.
For example, maximum position 1023 is 0x03FF in hexadecimal, or 0000 0011 1111 1111 in binary. The red portion represents the high byte H and the blue portion represents the low byte L.
Whenever a packet writes such a high/low pair, both bytes must be written together. Even if the high byte is zero, it must be filled with 0x00. For example, to write 1, write 0x0001.

Control Table Listing
EEPROM Region

Address	Name	Access	Initial value
0（0X00）	Model number (low byte)	R	12（0x0C）
1（0X01）	Model number (high byte)	R	0（0x00）
2（0X02）	Firmware version	R	28(0x1c)
3（0X03）	ID	RW	1（0x01）
4（0X04）	Baud rate	RW	1（0x01）
5（0X05）	Response delay	RW	250(0xFA)
6（0X06）	Minimum angle (low byte)	RW	0（0x00）
7（0X07）	Minimum angle (high byte)	RW	0（0x00）
8（0X08）	Maximum angle (low byte)	RW	0（0x00）
9（0X09）	Maximum angle (high byte)	RW	0（0x00）
11（0X0B）	Temperature upper limit	RW	70（0x46）
12（0X0C）	Voltage lower limit	RW	90（0x5A）
13（0X0D）	Voltage upper limit	RW	160（0xA0）

14（0X0E）	Maximum torque (low byte)	RW	200（0xC8）
15（0X0F）	Maximum torque (high byte)	RW	3（0x00）
16（0X10）	Status return level	RW	2（0x02）
17（0X11）	Speed mode	RW	1（0x01）
18（0X12）	Torque-disable alarm	RW	4（0x24）

RAM Region

Address	Item	Access	Initial value
24（0X18）	Torque enable	RW	0（0x00）
25（0X19）	Calibration enable	RW	0（0x00）
26（0X1A）	CW compliance margin	RW	0（0x00）
27（0X1B）	CCW compliance margin	RW	0（0x00）
28（0X1C）	CW compliance slope	RW	32（0x20）
29（0X1D）	CCW servo slope	RW	32（0x20）
30（0X1E）	Target position (low byte)	RW	Value at address 36
31（0X1F）	Target position (high byte)	RW	Value at address 37
32（0X20）	Moving speed (low byte)	RW	0
33（0X21）	Moving speed (high byte)	RW	0
34（0X22）	Torque limit (low byte)	RW	Value at address 14
35（0X23）	Torque limit (high byte)	RW	Value at address 15
36（0X24）	Current position (low byte)	R	
37（0X25）	Current position (high byte)	R	
38（0X26）	Current speed (low byte)	R	
39（0X27）	Current speed (high byte)	R	
40（0X28）	Current current (low byte)	R	
41（0X29）	Current current (high byte)	R	
42（0X2A）	Current voltage	R	
43（0X2B）	Current temperature	R	
44（0X2C）	Registered instruction flag	R	0（0x00）
46（0X2E）	Moving flag	R	0（0x00）
47（0X2F）	Flash region lock	RW	0（0x00）
48（0X30）	Collision value (low byte)	RW	32（0x20）
49（0X31）	Collision value (high byte)	RW	0（0x00）

Memory Address Descriptions

Address 0x03: Motor ID

The ID is the unique identifier of each motor, allowing instruction packets to distinguish between motors connected to the same bus.

Users can set motor IDs from 0 to 253 (0xFD); 254 (0xFE) is the designated broadcast ID.

Broadcast ID 0xFE can send an instruction simultaneously to every motor connected to the same bus.

Note: Motors on the same bus must not have duplicate IDs; otherwise, their communication packets will interfere with one another.

Address 0x04: Baud Rate

The baud rate determines communication speed. Formula: Speed (BPS) = 2,000,000 / (value at address 4 + 1).

Value at address 4	Hex	Calculated BPS	Actual BPS	Error
1	0x01	1000 000	1000 000	0.000%
3	0x03	500 000	500 000	0.000%
4	0x04	400 000	400 000	0.000%
7	0x07	250 000	250 000	0.000%
9	0x09	200 000	200 000	0.000%
16	0x10	117 647.1	115200.0	-2.124%
34	0x22	57142.9	57600.0	0.794%
103	0x67	19230.8	19200.0	-0.160%
207	0xCF	9615.4	9600.0	-0.160%

Address 0x05: Response Delay

This sets the delay between receipt of an instruction packet and the beginning of the response packet. Unit: 2 µs; range: 0–250.
For example, at the default setting of 250 (500 µs), the motor waits 500 microseconds after receiving the instruction packet before responding.

Addresses 0x06–0x09: Position Limits
These values set the minimum and maximum target positions within the range 0–1023.
Note: The minimum position limit must not exceed the maximum position limit.

Operating mode	**CW / CCW**
Wheel mode	Both equal 0
Joint mode	Not both 0

When both CW and CCW values are 0, the motor enters wheel mode.

Address 0x0B: Upper Temperature Limit
This sets the maximum operating temperature. Unit: 1°C; range: 0–100.
Note: After a temperature alarm, allow 20 minutes for cooling before use. Continuous operation at high temperatures can seriously damage the product.

Addresses 0x0C–0x0D: Voltage Limit Range
These values set the upper and lower motor operating voltage limits. Unit: 0.1 V; range: 50–255.

Addresses 0x0E–0x0F and 0x22–0x23:
These addresses limit the motor's maximum torque output.
The maximum torque can be set at two sets of addresses: EEPROM (0x0E–0x0F) and RAM (0x22–0x23).
After power-on, RAM values at 0x22–0x23 are initialized from EEPROM values at 0x0E–0x0F. The motor's final output is limited by the RAM values at 0x22–0x23.
Note: These registers are not used in your company's current application.

Address 0x10: Status Return Level
This value determines when the motor responds with a status packet after receiving an instruction packet.

Setting	Response condition	Description
0	Ping instruction	Motor responds only to Ping, not to other instructions.
1	Read and Ping instructions	Motor responds only to Read and Ping instructions.
2	All instructions	Motor responds to all instructions.

Note: When the instruction packet uses the broadcast ID, the motor does not send a response for any instruction.

Address 0x11: Speed Mode Setting
This register selects open-loop or closed-loop motor control.
When this value is 0, the motor uses open-loop control.
When this value is 1, the motor uses closed-loop control.

Address 0x12: Torque Shutdown Setting
If the corresponding bit is set to 1, the motor disables torque when the associated error occurs.

Bit	Function
7	0
6	If 1, disable torque on an instruction error.
5	If 1, disable torque on an overload error.
4	If 1, disable torque on a checksum error.
3	If 1, disable torque when the packet length exceeds the specified limit.
2	If 1, disable torque on an overheating error.
1	If 1, disable torque on an angle-limit error.
0	If 1, disable torque on an input-voltage error.

This function uses bitwise OR logic across the specified bits. Unlike an LED alarm, torque remains disabled even after the motor returns to normal status. To restore torque, set "Torque Enable" (address 0x18) to 1.
Note: This register setting is currently ineffective.

Address 0x18: Torque Switch
This value controls motor torque enable.
The default value at motor power-on is 0.

Value	Description
0	Disable torque; the motor produces no torque and the output shaft can be moved freely.
1	Enable torque; the motor operates according to its current operating mode.

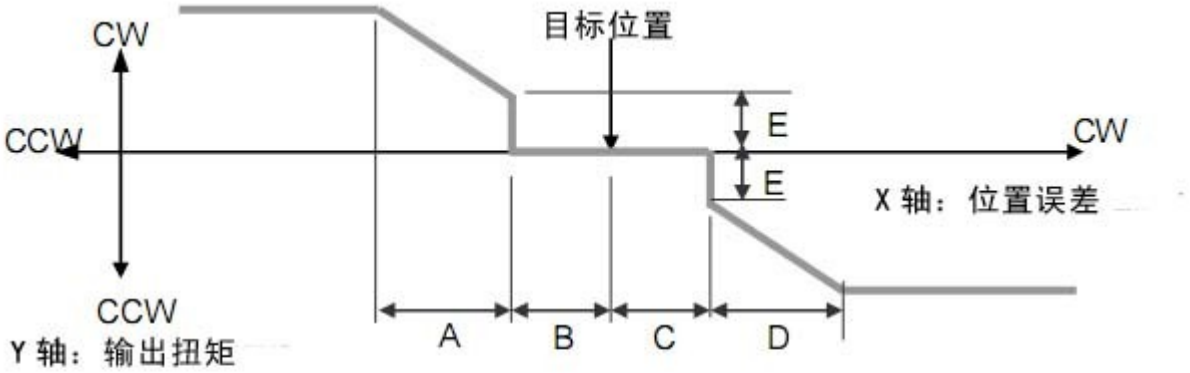
Address 0x19: Calibration Enable

This register controls motor calibration. Writing any parameter to it calibrates the motor, including phase current, DC-bus current, and electrical angle.

Note: Calibration is already performed at the factory; repeated calibration is unnecessary. If recalibration is still required, ensure the motor is unloaded because calibration forcibly moves the motor to a known position.

Addresses 0x1A–0x1D: Compliance Margins and Slopes

Motor compliance is represented by compliance margins and slopes. This feature can reduce shock to the output shaft. The following figure illustrates how each compliance region is represented by position error and output torque.

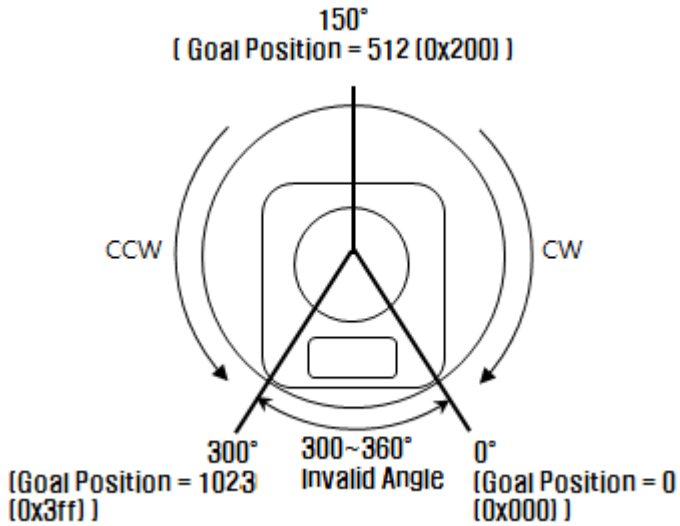


A：CCW 柔性斜线（地址 0x1D）
B：CCW 柔性边距（地址 0x1B）
C：CW 柔性边距边距（地址 0x1A）
D：CW 柔性斜线（地址 0x1C）
E：撞击（地址 0x30、0x31）

Note: These registers are not used in your company's current application.

Addresses 0x1E–0x1F: Target Position

The motor's physical operating range is 0–300°. The register value range is 0–1023 (0x3FF), with a unit of 0.29°. As illustrated:



Note: This value is invalid in wheel mode.

Note: Your company currently uses speed mode; these registers are not used in the current application.

Addresses 0x20–0x21: Moving Speed

This register controls motor speed, in °/s. It is an INT16 value; positive and negative signs indicate opposite directions. For example, −300°/s is 0xfed4.

Addresses 0x24–0x25: Current Motor Angular Position.
Addresses 0x26–0x27: Current Motor Speed.
Addresses 0x28–0x29: Current Motor Current.
Address 0x2A: Current Motor Voltage.
Address 0x2B: Current Motor Temperature.
Address 0x2C: 1 when an instruction is assigned using REG_WRITE; 0 when ACTION completes the assigned instruction.
Address 0x2E: 1 when the motor is moving. (Note: This register is not used.)
Address 0x2F: Flash Lock. If set to 1, the Flash region cannot be written. Once locked, it can only be unlocked by switching off power and restarting the motor.
Addresses 0x30–0x31: Collision Value, representing the minimum current used during motor operation. Initial value: 0x20; maximum: 0x3ff. (Note: This register is ineffective.)

Figure 1 — Compliance characteristic diagram: English legend

Vertical axis: Output torque. Horizontal axis: Position error. Target position is marked on the graph. CW = clockwise; CCW = counterclockwise.

A: CCW compliance slope (address 0x1D). B: CCW compliance margin (address 0x1B). C: CW compliance margin (address 0x1A). D: CW compliance slope (address 0x1C). E: Collision/impact threshold (addresses 0x30 and 0x31).

Original note: These registers are not used in your company’s current application.

Figure 2 — Position range diagram: English legend

Top: 150° [Goal Position = 512 (0x200)]. Left: 300° [Goal Position = 1023 (0x3FF)]. Right: 0° [Goal Position = 0 (0x000)]. Bottom: 300°–360° is an invalid angle range. Arrows indicate CCW and CW directions.

Original notes: This value is invalid in wheel mode. The current application uses speed mode, and these position registers are not used.